

THE LASER-RAMAN SPECTRUM OF A DAVYDOV SOLITON

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It is suggested that the laser-Raman spectra between 30 and 300 cm^{-1} , which have recently been observed on metabolically active cells, may reflect the internal vibrations of Davydov solitons.

A recent survey of laser-Raman spectroscopy of living cells [1] reveals, among many others, the following facts: (i) at 300 K a Raman spectrum is observed only when cells are metabolically active, (ii) the intensity ratios of Stokes to anti-Stokes lines indicate that the Raman active states are produced *in vivo* by non-thermal means, and (iii) spectral lines below 300 cm^{-1} shift to lower wave numbers as the cells progress through their life cycles. A particular spectrum of *E. coli* (taken from fig. 7 of ref. [1]) shows between 30 and 200 cm^{-1} the lines mentioned in table 1.

The purpose of this note is to point out the spectrum of table 1 may reflect internal vibrations of the alpha-helix soliton introduced by Davydov [2]. It has recently been suggested that the phase changes observed from the Raman spectra of green algae around 260 K [3] may be influenced by the presence of Davydov solitons [4].

Previous numerical studies [5] have shown that the anharmonic character of a real alpha-helix has the

proper order of magnitude to support Davydov solitons. Recently this numerical model has been made more precise through the following modifications: (i) ten additional dipole-dipole coupling terms have been included to make the threshold for soliton formation insensitive to the number of such terms, (ii) the dynamical equations were modified to reflect the true helical symmetry, and (iii) the linear restoring force of the longitudinal hydrogen bonds was taken to be the value (21 N/m) obtained by Kuprievich and Kudritskaja from SCF calculations on formamide dimer [6]. A critical nonlinear parameter in the model is χ_1 , the change in energy of an amide-I bond per unit extension of a corresponding longitudinal hydrogen bond.

A summary of numerical results is displayed in fig. 1. Here the ordinate (bond energy) is the amide-I occupation probability plotted against unit cell number at a particular value of time for various values of χ_1 between 0 and 0.8×10^{-10} N. It is interesting to note that a soliton forms at just the value ($\chi_1 = 0.34 \times 10^{-10}$ N) that Kuprievich and Kudritskaja have computed for the hydrogen bond of formamide [6]. To appreciate the internal dynamics of a soliton consider fig. 2. Here, in addition to total amide-I occupation probability at each unit cell, are plotted also components along the three "spines" (etc. $\text{N}-\text{C}=\text{O} \cdots \text{N}-\text{C}=\text{O} \cdots$ etc.) that run longitudinally along the helix. An inter-spine oscillation is clearly observed. At computer time $\tau = 150$ and 152, spine #1 carries more energy; at $\tau = 156-162$, spine #2 carries more; and so forth. The cycle repeats after twenty units of computer time

Table 1

Line	cm^{-1}
1	45
2	52
3	63
4	85
5	90
6	108
7	123
8	152
9	182

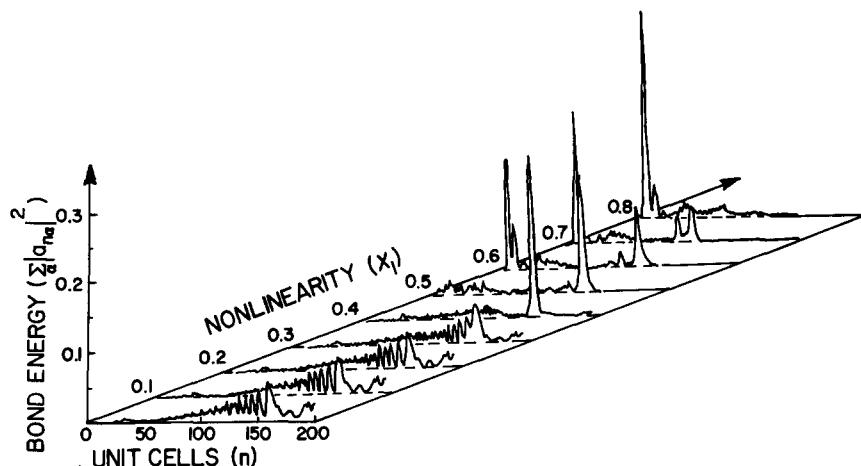


Fig. 1. Plots of bond occupation probability versus unit cell for several values of the nonlinearity parameter χ_1 ($\times 10^{-10}$ N) at computer time $\tau = 400$. The computation was begun with one quantum in each of the first two unit cells at $\tau = 0$.

which corresponds to a physical time $T_1 = 2 \times 10^{-12}$ s. A shorter oscillation, also seen from fig. 2, is caused by interaction of the soliton with the fundamental periodicity of the unit cell. The period of this oscillation is

$$T_2 = \frac{8}{3} \times 10^{-13} \text{ s}.$$

Raman lines should be expected at energies

$$E_i = 1/cT_i \text{ cm}^{-1},$$

corresponding to these two periods and their sums and differences. Since

$$E_1 = 17 \text{ cm}^{-1}, \quad E_2 = 125 \text{ cm}^{-1},$$

table 2 is readily constructed.

Comparison of the measured lines in table 1 with those calculated in table 2 reveals a striking similarity. (It should be emphasized that the periods T_1 and T_2 were computed long before the author was aware of

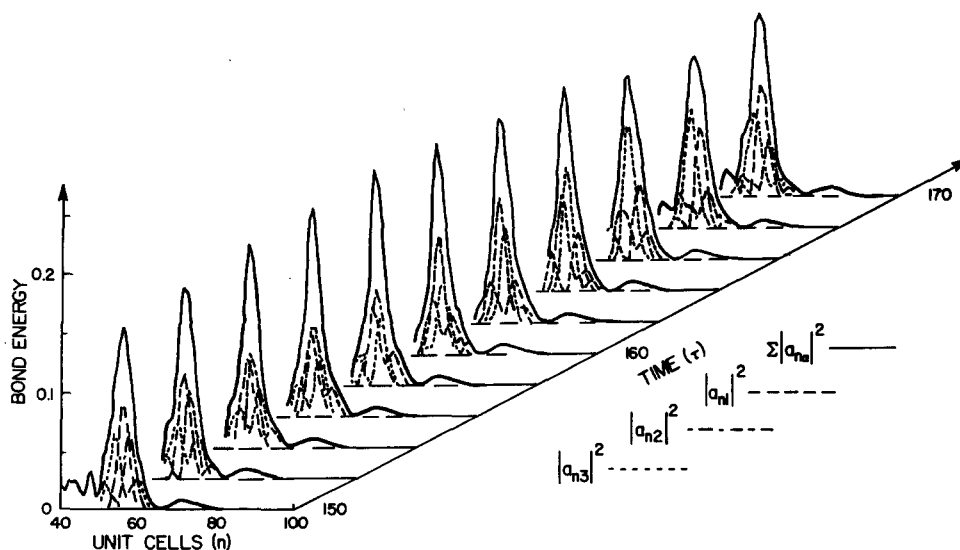


Fig. 2. Sketch of numerical calculations showing the internal dynamics of an alpha-helix soliton. The initial conditions are as in fig. 1 and $\chi_1 = 0.4 \times 10^{-10}$ N.

Table 2

Line	Structure	cm^{-1}
1	$2E_1$	34
2	$3E_1$	51
3	$4E_1$	68
4	$5E_1$	85
5	$E_2 - 2E_1$	91
6	$E_2 - E_1$	108
7	E_2	125
8	$E_2 + 2E_1$	159
9	$E_2 + 3E_1$	176

ref. [1].) Furthermore the tendency of such spectral lines to shift to lower wave numbers as cells progress through their life cycles is gracefully explained by assuming that solitons move more slowly as a cell ages.

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